

2026 March US II

Module 1

Question 1

$$fx = 23(1.20)^{\frac{x}{8}}$$

For the given function f , the value of fx increases by $p\%$ for every increase of x by 8. What is the value of p ?

- A) 20
- B) 31
- C) 40
- D) 44

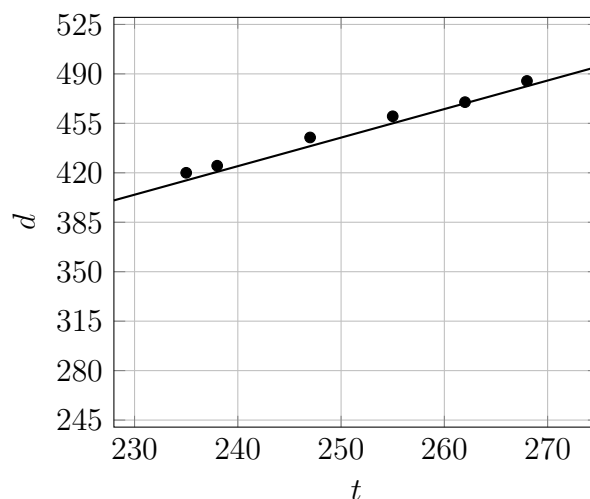
Question 2

A model estimates that in a particular forest, the number of trees with any given diameter measured at shoulder height is 21% less for each 1-inch increase in tree diameter measured at shoulder height. The model can be written in the form $fx = ab^x$, where a and b are constants and x is the tree's diameter, in inches, measured at shoulder height, and $x \geq 5$. The model estimates that 3,100 trees in this forest have a diameter of 13 inches measured at shoulder height. Which function best represents this model?

- A) $fx = 3,100 \cdot 0.21^x$
- B) $fx = 3,100 \cdot 0.79^x$
- C) $fx = 66,000 \cdot 0.21^x$
- D) $fx = 66,000 \cdot 0.79^x$

Question 3

The scatterplot shows the relationship between two variables, t and d .



Which of the following equations is the most appropriate linear model for the data shown?

- A) $d = -60.1 + 2.02t$
- B) $d = 160.1 + 2.02t$
- C) $d = 359.8 + 2.02t$
- D) $d = 394.8 + 2.02t$

Question 4

The function f is defined by $f(x) = a^x - b$, where a and b are constants. In the xy -plane, the graph of $y = fx$ passes through the points $(c, 11)$ and $(2c, 221)$, where c is a constant. Which of the following could be the value of b ?

- A) 4
- B) 25
- C) 210
- D) 232

Question 5

In triangle ABC , the measure of angle A is 56° and $AC = 30$. In triangle PQR , the measure of angle P is 56° and $PR = 90$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR ?

- A) $AB = 20$ and $PQ = 20$.
- B) $AB = 20$ and $QR = 60$.
- C) The measures of angle B and angle R are 48° and 76° , respectively.
- D) The measures of angle B and angle Q are 56° and 48° , respectively.

Question 6

$$\frac{5}{9}(5x + 9)(x + \sqrt{5k + 9})(x - \sqrt{5k + 9}) = 0$$

In the given equation, k is a positive constant. The product of the solutions to the equation is 81. What is the value of k ?

Question 7

$$4x^2 - px + w = -86$$

In the given equation, p and w are integer constants. The equation has exactly one real solution. Which is **NOT** a possible value of w ?

- A) -22
- B) 14
- C) 25
- D) 314

Question 8

If $\frac{x+6}{5} = \frac{x+6}{-13}$, the value of $x+6$ is between which of the following pairs of values?

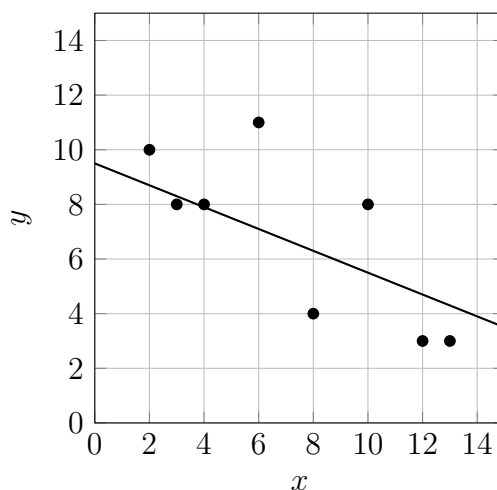
- A) -7 and -5
- B) -2 and 2
- C) 2 and 7
- D) 8 and 13

Question 9

The expression $6x^4 + 17x^2 + 5$ can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers, or as $(3x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. What is the value of $a + c$?

Question 10

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



Which of the following equations best represents the line of best fit shown?

- A) $y = 9.5 + 0.4x$
- B) $y = 9.5 - 0.4x$
- C) $y = -9.5 + 0.4x$
- D) $y = -9.5 - 0.4x$

Question 11

The graph of the polynomial function f in the xy -plane, where $y = f(x)$, passes through the point $(9, 8)$. Which of the following must be true?

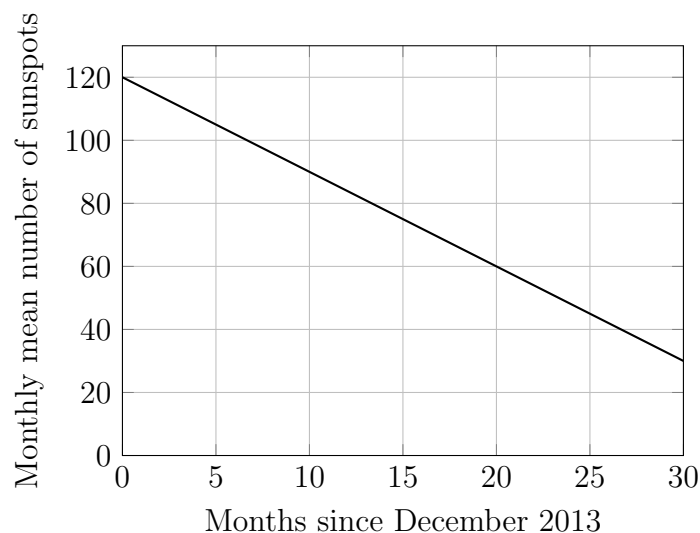
- A) $f(0) = 8$
- B) $f(8) = 9$
- C) $f(9) = 0$
- D) $f(9) = 8$

Question 12

For the exponential function g , the value of $g(x)$ doubles for every increase of 2 in the value of x . If $g(4) = 9$, what is the value of $g(6)$?

Question 13

Sunspots are temporary dark spots on the surface of the Sun. The monthly mean number of sunspots for one calendar month is the average of the number of sunspots observed each day during the month. The linear function s models the monthly mean number of sunspots as a function of the number of months, x , since December 2013, where $0 \leq x \leq 30$. The graph of $y = s(x)$ is shown.



Which of the following is the best estimate for the monthly mean number of sunspots in December 2014?

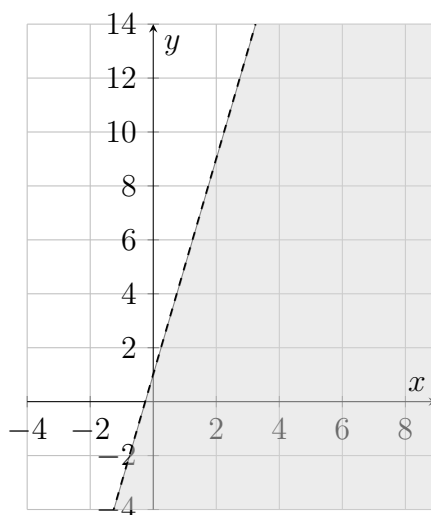
- A) 61
- B) 71
- C) 81
- D) 91

Question 14

What is the x -intercept of a line in the xy -plane that has a slope of $-\frac{4}{7}$ and passes through the point $(0, 12)$?

- A) $(-21, 0)$
- B) $(-4, 0)$
- C) $(7, 0)$
- D) $(21, 0)$

Question 15



The shaded region shown represents the solutions to which inequality?

- A) $y < \frac{1}{4}x + 1$
- B) $y < 4x + 1$
- C) $y > \frac{1}{4}x + 1$
- D) $y > 4x + 1$

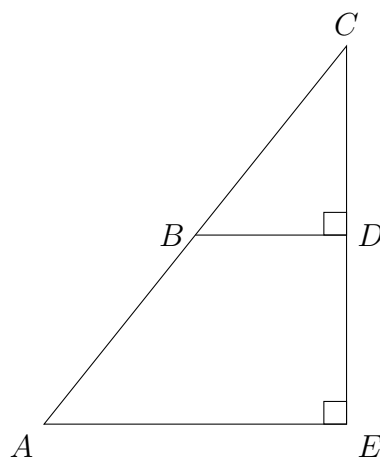
Question 16

$$y = 5x + 18$$

$$y = 5x - 18$$

How many solutions does the given system of equations have?

- A) Zero
- B) Exactly one
- C) Exactly two
- D) Infinitely many

Question 17

Note: Figure not drawn to scale.

If $\cos A = 0.65$, what is the value of $\sin C$?

- A) 0.87
- B) 0.65
- C) 0.35
- D) 0.13

Question 18

The function f is defined by $f(x) = 3x - 7$ and $f(b) = \frac{4}{5}$. What is the value of b ?

Question 19

$$9|7 - x| + 5 = 95$$

What is the sum of the solutions to the given equation?

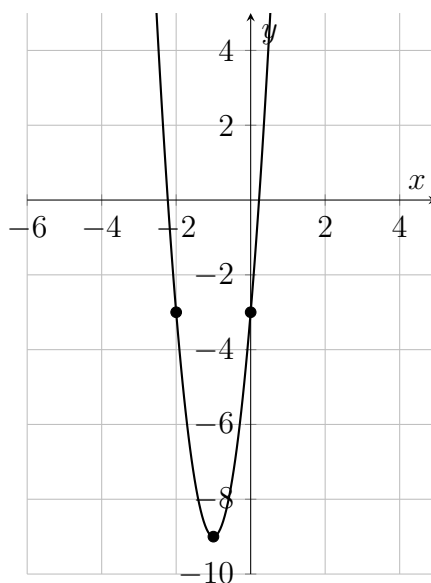
- A) 14
- B) 10
- C) -3
- D) -14

Question 20

$$f(x) = x^2 - 4x - 780$$

The function f is defined by the given equation. Which of the following equivalent forms of the equation displays the minimum value of the function as a constant or coefficient?

- A) $f(x) = x^2 - 4x + 195$
- B) $f(x) = (x - 2)^2 + (-784)$
- C) $f(x) = x(x - 4) + (-780)$
- D) $f(x) = (x + 26)(x - 30)$

Question 21

The graph of $y = 6x^2 + bx + c$ is shown, where b and c are constants. What is the value of bc ?

Question 22

The value of a painting increased by 166% from the end of 2011 to the end of 2012 and then decreased by 14% from the end of 2012 to the end of 2013. What was the net percentage increase in the value of the painting from the end of 2011 to the end of 2013?

- A) 128.76%
- B) 142.76%
- C) 152.00%
- D) 203.24%

Module 2

Question 1

An object's speed is increasing at a rate of 12.60 meters per second squared. What is this rate, in miles per minute squared, rounded to the nearest tenth? (Use 1 mile = 1,609 meters.)

Question 2

A chemist mixed x liters of a 6% saline solution with y liters of a 9% saline solution to produce a 7% saline solution. Which equation best represents this situation? (Assume the volumes of the solutions are additive.)

- A) $0.06x + 0.09y = 7(x + y)$
- B) $0.06x + 0.09y = 0.07(x + y)$
- C) $0.6x + 0.9y = 7(x + y)$
- D) $0.6x + 0.9y = 0.7(x + y)$

Question 3

The expression $(3x + 5)(7x - 8)$ can be written in the form $ax^2 + bx + c$, where a , b , and c are constants. What is the value of $a + b$?

Question 4

A rectangle has a length of x units and a width of $x - 3$ units. If the rectangle has an area of 28 square units, what is the value of x ?

- A) 4
- B) 7
- C) 11
- D) 28

Question 5

A number x is at most 27 less than 3 times the value of y . If the value of y is 8, what is the greatest possible value of x ?

Question 6

$$\frac{1}{x-3} = \frac{x-1}{-x+1.75}$$

Which of the following is a solution to the given equation?

- A) 0.75
- B) 2.5
- C) 3
- D) 4

Question 7

Number of cars	Maximum number of passengers and crew
2	81
5	189
10	369

The table shows the linear relationship between the number of cars, c , on a commuter train and the maximum number of passengers and crew, p , that the train can carry. Which equation represents the linear relationship between c and p ?

- A) $36c - p = -9$
- B) $36c - p = 9$
- C) $36p - c = -9$
- D) $36p - c = 9$

Question 8

$$20x = 1,200y - 2,000$$

One of the two equations in a system of linear equations is given. The system has no solution. Which equation could be the second equation in this system?

- A) $\frac{1}{20}x = 3y$
- B) $\frac{1}{20}x = 60y - 100$
- C) $x = 3y$
- D) $x = 60y - 100$

Question 9

A circle has center G , and points M and N lie on the circle. Line segments MH and NH are tangent to the circle at points M and N , respectively. If the radius of the circle is 247 millimeters and the perimeter of quadrilateral $GMHN$ is 5,174 millimeters, what is the distance, in millimeters, between points G and H ?

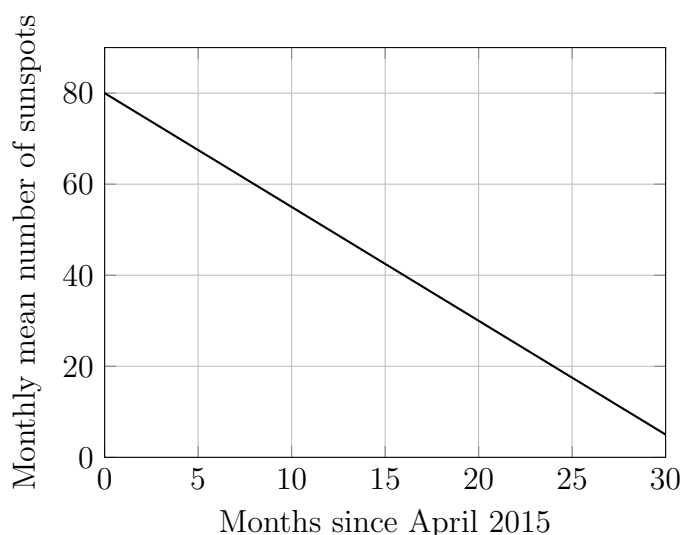
Question 10

In triangle JKL , the measure of angle J is $90b^\circ$, the measure of angle K is $66a^\circ$, and the measure of angle L is $24a^\circ$, where a and b are constants. Which of the following must be true?

- A) $\cos L > \sin K$
- B) $\cos L = \sin K$
- C) $\cos L < \sin K$
- D) There is not enough information to compare the values of $\cos L$ and $\sin K$.

Question 11

Sunspots are temporary dark spots on the surface of the Sun. The monthly mean number of sunspots for one calendar month is the average of the number of sunspots observed each day during the month. The linear function s models the monthly mean number of sunspots as a function of the number of months, x , since April 2015, where $0 \leq x \leq 30$. The graph of $y = sx$ is shown.



Which of the following is the best estimate for the monthly mean number of sunspots in April 2016?

- A) 23

B) 33

C) 43

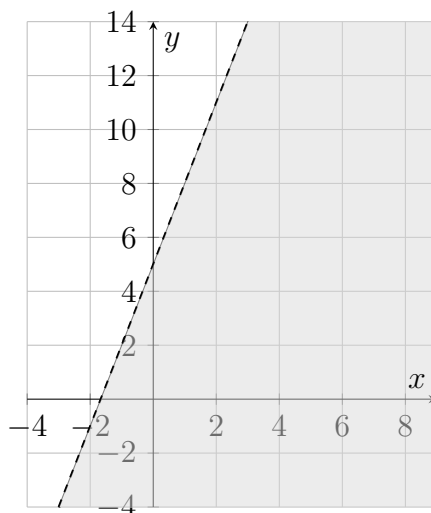
D) 53

Question 12

For the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of -4 and passes through the point $(3, -5)$. Which equation defines f ?

A) $f(x) = 3x - 5$ B) $f(x) = -4x - 5$ C) $f(x) = -4x - 4$ D) $f(x) = -4x + 7$

Question 13



The shaded region shown represents the solutions to which inequality?

A) $y < \frac{1}{3}x + 5$ B) $y < 3x + 5$ C) $y > \frac{1}{3}x + 5$ D) $y > 3x + 5$

Question 14

A circle has a radius of 4.3 inches. The area of the circle is $b\pi$ square inches, where b is a constant. What is the value of b ?

Question 15

$$y \leq x + 8$$

$$y \geq -3x - 8$$

Which point (x, y) is a solution to the given system of inequalities in the xy -plane?

- A) $(0, -9)$
- B) $(0, 9)$
- C) $(-5, 0)$
- D) $(5, 0)$

Question 16

If $3\sqrt{3} + 6\sqrt{27} - \sqrt{48}$ can be rewritten as $x\sqrt{3}$, then what is the value of x ?

Question 17

$$7|6 - x| + 3 = 73$$

What is the sum of the solutions to the given equation?

- A) 12
- B) 10
- C) -4
- D) -12

Question 18

$$\frac{8cx + 7}{5} = 4x - 7$$

In the given equation, c is a constant. If the equation has no solution, what is the value of c ?

- A) $\frac{5}{2}$
- B) $\frac{2}{5}$
- C) $-\frac{2}{5}$
- D) $-\frac{5}{2}$

Question 19

The function f is defined by $f(x) = 3x - 7$. If $f(a + 1) = 2a$, what is the value of a ?

- A) -1
- B) 1
- C) 5
- D) 6

Question 20

A certain investment account offers a special interest rate for the first 4 months the account is open followed by a lower interest rate for the remainder of the time the account is open. Bennett opened one of these accounts with an original account balance of \$700 and did not make any other deposits or withdrawals. 4 months after Bennett opened the account, the balance had increased by 0.6% of the original balance. 6 months after Bennett opened the account, the balance had increased by an additional 0.2% of the balance at the end of the first 4 months. Every 2 months after the first 6 months, the balance had increased by an additional 0.2% of the balance 2 months before. Which of the following equations could represent the account balance Bx , in dollars, x months after the account was opened, where $x \geq 4$?

- A) $Bx = 704.20 \cdot 1.002^{\frac{x}{2} - \frac{x}{2}}$
- B) $Bx = 704.20 \cdot 1.002^{\frac{x}{2} - 4}$
- C) $Bx = 704.20 \cdot 1.002^{2x-8}$
- D) $Bx = 704.20 \cdot 1.002^{2x-4}$

Question 21

A circle in the xy -plane has its center at $(-5, 5)$. Line t is tangent to this circle at the point $(6, -1)$. Which of the following points also lies on line t ?

- A) $\left(0, \frac{11}{6}\right)$
- B) $(1, 16)$
- C) $(12, 10)$
- D) $(17, 5)$

Question 22

$$4x^2 - px + w = -85$$

In the given equation, p and w are integer constants. The equation has exactly one real solution. Which is **NOT** a possible value of w ?

- A) -21
- B) 15
- C) 64
- D) 315

Answer Sheet

Module 1

1	2	3	4	5	6	7	8	9	10
<i>D</i>	<i>D</i>	<i>A</i>	<i>A</i>	<i>C</i>	36/5	<i>C</i>	<i>B</i>	17/2	<i>B</i>

11	12	13	14	15	16	17	18	19	20
<i>D</i>	18	<i>D</i>	<i>D</i>	<i>B</i>	<i>A</i>	<i>B</i>	13/5	<i>A</i>	<i>B</i>

21	22
-36	<i>A</i>

Module 2

1	2	3	4	5	6	7	8	9	10
28.2	<i>B</i>	32	<i>B</i>	-3	<i>B</i>	<i>A</i>	<i>A</i>	2353	<i>D</i>

11	12	13	14	15	16	17	18	19	20
<i>D</i>	<i>D</i>	<i>B</i>	18.49	<i>D</i>	17	<i>A</i>	<i>A</i>	<i>D</i>	<i>A</i>

21	22
<i>C</i>	<i>C</i>